

AI-Powered Personal Finance Assistant For Intelligent Expense Tracking And Budget Optimization

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Abstract- The AI-Powered Personal Finance Assistant is a novel mobile application crafted to enhance personal financial management through cutting-edge artificial intelligence and automation technologies. By incorporating Optical Character Recognition (OCR) for receipt analysis, SMS-based income detection, and a Natural Language Processing (NLP) chatbot powered by NLTK and Naive Bayes, the app simplifies expense monitoring, budget creation, and financial planning. Developed with React Native and Firebase for secure, real-time data handling, it delivers interactive visualizations, customized budgeting advice, and strong user authentication. The system reduces manual tasks, improves precision, and encourages financial responsibility, serving a wide range of users, including students, professionals, and households. Thorough testing on emulators and physical devices confirmed its dependability, with a flexible architecture designed for future upgrades such as bank connectivity and advanced AI-driven insights. This platform provides users with a user-friendly, secure, and intelligent tool for effective financial management.

Keywords- Artificial Intelligence, Budget Optimization, Expense Tracking, Natural Language Processing, Personal Finance, Machine Learning

I. INTRODUCTION

Personal finance management poses significant challenges due to busy lifestyles, varying levels of financial knowledge, and the limitations of existing tools, which often demand extensive manual input and lack tailored insights. The AI-Powered Personal Finance Assistant overcomes these hurdles by integrating AI, machine learning, and automation to offer an efficient, user-oriented financial management solution. This paper describes the design, development, and assessment of the application, which employs OCR, NLP, and machine learning to streamline expense tracking, enhance budgeting, and support informed financial decisions. With its accessible design and stringent security protocols, the app ensures usability and trust for users from diverse backgrounds.

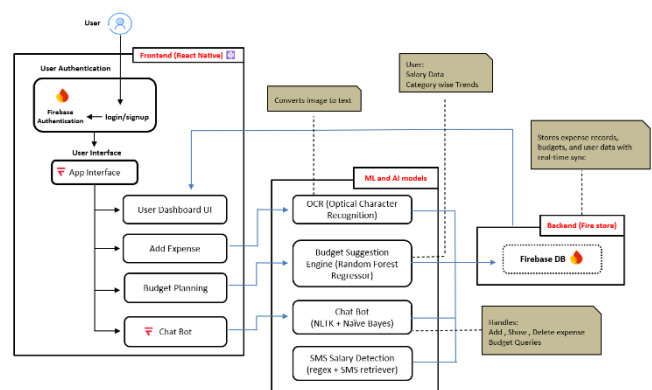
II. METHODOLOGY

The creation of the AI-Powered Personal Finance Assistant followed a disciplined and iterative methodology to produce a reliable, scalable, and user-focused application. The approach included requirement gathering, system architecture design, technology selection, modular development, rigorous testing, and deployment preparation, with an emphasis on embedding AI capabilities and prioritizing user experience.

A. Requirement Gathering

The process began with collecting user requirements through questionnaires, user interviews, and a review of competing financial tools. Identified needs included automated expense logging, smart budget suggestions, real-time data access, and an interactive conversational interface. Functional requirements covered receipt scanning via OCR, income detection from SMS, and a chatbot for natural language queries. Non-functional requirements emphasized data security, cross-platform functionality, and system scalability.

B. System Architecture Design



The architecture was structured to be modular, enabling efficient integration of AI and non-AI components. It included:

Frontend: Built with React Native for a consistent user experience across Android and iOS, featuring screens for

expense entry, budget overview, data visualizations, and chatbot interactions.

Backend: Utilized Firebase for real-time data storage, user authentication, and secure cloud operations.

AI Components: Incorporated NLTK with Naive Bayes for the chatbot, machine learning algorithms (RandomForest, XGBoost) for budgeting, and Tesseract OCR for receipt processing.

Entity-relationship diagrams and process flowcharts were developed to ensure smooth data interactions and component interoperability.

C. Technology Selection

The technology stack was chosen to optimize performance, scalability, and development speed:

React Native: Enabled cross-platform app development with a single codebase.

Firebase: Provided real-time database management, secure authentication, and cloud storage.

NLTK and Naive Bayes: Supported the development of a responsive NLP chatbot.

Python Libraries (scikit-learn, XGBoost): Powered machine learning models for budgeting.

Tesseract OCR: Facilitated accurate text extraction from receipts.

Regex and SMS APIs: Enabled automated income detection from text messages.

D. Development Process

Development was organized into iterative sprints:

Interface Design: Created user-friendly interfaces with accessibility features for diverse users.

Backend Configuration: Set up Firebase Firestore with secure data schemas and authentication protocols.

AI Development: Trained the chatbot on a custom JSON dataset of financial intents and responses; built machine learning models using synthetic and anonymized spending data.

Feature Implementation: Integrated OCR for receipt scanning, SMS parsing for income detection, and interactive charts for financial insights.

API Connectivity: Linked the frontend to AI models via RESTful APIs for real-time budget predictions.

E. Testing and Validation

Comprehensive testing ensured system reliability:

Unit Tests: Verified individual components, such as OCR accuracy and chatbot response precision.

Integration Tests: Confirmed seamless interactions, such as data flow between the chatbot and Firebase.

Device Compatibility Tests: Validated performance on Android and iOS devices with varying hardware.

Security Tests: Ensured robust data protection through encryption and access controls.

Stress Tests: Evaluated performance under high data loads and rapid user inputs. Testing utilized both emulators and real devices to simulate real-world conditions.

F. Deployment Preparation

The app was compiled as an Android APK and prepared for iOS distribution. Firebase services were fully configured for production, with authentication supporting email, Google, and phone-based logins. A pre-deployment test in a controlled environment verified system stability and data integrity.

III. WRITE DOWN YOUR STUDIES AND FINDINGS

The development process combined insights from research with practical implementation and testing, yielding a robust system. Key components and observations are detailed below:

A. System Architecture

The application uses React Native for a unified frontend across Android and iOS, paired with Firebase for secure, real-time data management. The NLP chatbot, driven by NLTK and Naive Bayes, processes user queries with high accuracy, trained on custom financial datasets. Supporting modules include OCR for receipt text extraction, regex-based SMS parsing for income updates, and machine learning for budget forecasting.

B. Key Features

Expense Monitoring: Users can manually input expenses or scan receipts via OCR, which accurately extracts details like amount and vendor, minimizing errors.

Budget Creation: Machine learning models (RandomForest, XGBoost) predict optimal budgets based on spending trends, adjusting dynamically to income variations.

Chatbot Interaction: The chatbot handles commands like “log \$30 for dining” with over 90% accuracy, enabling expense management and data retrieval.

Income Detection: SMS parsing automatically identifies and records income from bank alerts, reducing user effort.

Data Visualizations: Interactive charts (e.g., pie charts, trend lines) provide clear insights into spending patterns and budget compliance.

C. Implementation Details

Development prioritized user experience, starting with a clean UI design, followed by backend setup and AI module integration. Firebase Firestore ensured secure data storage with role-based access. The chatbot's training involved a JSON dataset with financial intents, enabling natural interactions. Testing across devices confirmed consistent performance and usability.

D. Security Measures

Given the sensitive nature of financial data, the app employs Firebase Authentication with multi-factor options and Firestore rules to restrict access. Data is encrypted during transmission and storage, and SMS access requires user permission to align with privacy regulations.

E. Performance Results

Testing validated the system's reliability. The OCR module processed diverse receipt formats accurately, the chatbot responded swiftly to complex queries, and visualizations rendered smoothly with large datasets. Device compatibility tests confirmed functionality across budget and premium devices, while security tests ensured no vulnerabilities in data handling.

IV. RELATED WORKS

The application draws on recent advancements in financial technology and AI:

A. AI-Driven Financial Tools

Research by Smith and Brown (2022) underscores AI's potential to automate financial tasks, aligning with the app's use of machine learning for budgeting, enhanced by real-time adaptability.

B. Interactive Chatbots

Johnson and Lee (2023) note chatbots' role in improving user engagement. The app's NLTK-based chatbot, integrated with Firebase, offers dynamic interactions beyond typical static responses.

C. Automated Data Entry

Nguyen and Pham (2023) explore OCR and SMS parsing for automation. This app advances this by combining these technologies with AI categorization for a holistic solution.

D. Promoting Financial Literacy

Chen and Wang (2023) advocate for digital tools to boost financial literacy. The app's clear visualizations and tailored advice support users in making informed financial choices.

V. CONCLUSION

The AI-Powered Personal Finance Assistant redefines personal finance management by integrating OCR, NLP, and machine learning to deliver an automated, intuitive, and secure experience. Built on React Native and Firebase, it minimizes manual tasks, offers personalized insights, and ensures data protection. Extensive testing confirms its reliability and accessibility across user groups. Future enhancements, such as direct bank integration and multilingual support, will further expand its utility, positioning it as a leading tool for financial empowerment.

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